

Krutik Tikam

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SUMMARY

Analytical and results-oriented Full-Stack Developer and AI/ML Engineer with a robust background in building scalable web architectures, optimized backend routing, and predictive pipelines. Experienced in developing decoupled, async REST APIs, real-time analytics dashboards, and automated ETL workflows using Next.js 15, FastAPI, PostgreSQL, and Python. Passionate about engineering high-performance systems and deploying responsive, user-centric interfaces.

EDUCATION

Nagandas Khandwala College

Mumbai, Maharashtra

B.Sc. Computer Science (Spl. in Artificial Intelligence & Machine Learning)

2023 – 2026

- **Academic Standing:** CGPA: 7.5
- **Focus Area:** Data Structures & Algorithms, Database Management Systems, Neural Networks, Distributed Web Architecture.

PROFESSIONAL EXPERIENCE

Talent Corner HR Services Private Limited

Mumbai, India

Full-Stack Developer Intern

Dec 2025 – Feb 2026

- Developed and maintained responsive full-stack applications using **Python**, **FastAPI**, and frontend JavaScript frameworks, streamlining internal data management processes.
- Designed and deployed decoupled, production-ready **RESTful APIs** with asynchronous backend endpoints, optimizing server-side data processing cycles.
- Collaborated on database architecture design utilizing **PostgreSQL**, drafting complex SQL schemas and queries to drastically minimize search latency across high-volume applicant indexes.
- Integrated version control workflows using **Git/GitHub** and conducted rigorous API testing via Postman to lower application deployment friction.

TECHNICAL SKILLS

Frontend: React, Next.js 15 (App Router), TypeScript, JavaScript, Tailwind CSS, Recharts (D3.js), Framer Motion

Backend & Databases: FastAPI (Async), Node.js, Python, PostgreSQL (Supabase), SQL Alchemy, SQLite, Better-SQLite3, RESTful APIs

Data Analytics & ML: Pandas, NumPy, Scikit-Learn (K-Means, Cosine Similarity), XGBoost, ETL Pipelines, Vector Search (ChromaDB), Data Visualization, Jupyter

DevOps & Tools: Git, GitHub, Docker, Vercel, Render, Postman, VS Code, Tesseract.js, Lab Streaming Layer (LSL), AWS, Blender API

PROJECTS

Blender Robotic Arm Simulation | *Python, OpenAI Gymnasium, PPO, Blender API, Stable-Baselines3*

- Engineered a high-fidelity 3D robotic arm simulation environment in **Blender** utilizing closed-loop waypoint trajectory tracking and a proportional (P-control) feedback subsystem.
- Integrated an **OpenAI Gymnasium** environment wrapper to train continuous action-space reinforcement learning policies via Stable-Baselines3 **PPO** algorithms.
- Developed a smart raycast-based contact-sensing pipeline to detect exact finger alignment geometry, auto-locking joint rotations on impact to eliminate mesh penetration anomalies.
- Automated low-latency telemetry logging architectures to pipeline live joint coordinates, target vectors, and gripper metrics into a structured data analytics pipeline.

Football Any-latics Pro | *Python, XGBoost, Streamlit, Pandas, REST APIs*

- Designed and engineered an automated **ETL Pipeline** in Python that ingests, cleans, and structures live match fixtures from open sports REST APIs into highly optimized relational databases.
- Trained a predictive machine learning pipeline using **XGBoost** to compute live match win probabilities, achieving a high validation accuracy metric across dense sport datasets.

- Developed an interactive, web-based analytics dashboard using **Streamlit** to visualize real-time match projections, decreasing analytical processing overhead for complex sport metrics.

PokéArchitect: Full-Stack Strategy & Analytics Platform | *React, FastAPI, PostgreSQL (Supabase), Scikit-Learn*

- Developed a decoupled, high-performance web platform utilizing a robust **FastAPI** backend and a responsive **React/Vite** frontend secured with strict cross-origin hardening rules.
- Built an analytical heuristic intelligence engine leveraging **K-Means Clustering** for granular role discovery across 1,000+ data profiles and **Cosine Similarity** for real-time strategic recommendations.
- Designed rich meta-analytics layout modules using **Recharts (D3)** to track historical equilibrium, automating tabular database seeding workflows on application initialization.

OmniMath-Local: Asynchronous Verification & Vector RAG Platform | *FastAPI, ChromaDB, Pydantic, SymPy*

- Created an enterprise-grade async backend framework serving custom algorithmic validation logic and symbolic math verification workflows.
- Implemented semantic vector search using **ChromaDB** for a local RAG system, developing structured text chunking parameters for precise indexing within context windows.
- Optimized system throughput and minimized disk read/write boundaries by leveraging explicit asynchronous database workers and strict execution validation via **Pydantic schemas**.

CERTIFICATIONS

Cloud & Generative AI: AWS Completion Certificate: Data Engineering for Generative AI Applications, AWS Completion Certificate: Connecting Systems and Machines to AWS for Industrial Manufacturing, Microsoft Azure Cloud Computing (National Level SDP)

ML & Data Science: DeepLearning.AI Machine Learning Specialization, TensorFlow Developer Professional Certificate

Web Development: Meta Front-End Developer Professional Certificate, Responsive Web Design (freeCodeCamp)

INTERESTS

Data Analytics, Scalable Web Architectures, UI/UX Design, Open Source Contribution, Machine Learning Systems